

**Leader: corridor authority
(here and now)**

electrify edge $y_e \in \{0, 1\}$
charging bays g_e (350 kW each)
corridor charge θ_s per diesel truck-km
charging surcharge σ_s per kWh

y, g, θ, σ

**Followers: carrier coalition
(one programme per scenario)**

loaded flows, pooled empty repositioning
traction mode: electric or diesel
platoon formation and charging
linear programme $\Rightarrow$ exact duality

best response

solved as one MILP

Exact single-level reformulation

primal feasibility

$$Ax \geq b(w)$$

+

dual feasibility

$$A^\top \lambda \leq c(w)$$

+

strong duality

$$c(w)^\top x = b(w)^\top \lambda$$

every leader $\times$ follower product is binary $\times$ bounded-continuous and linearises exactly
no complementarity binaries; the one dual bound the model needs is proved by an exchange argument, not chosen